Class Imbalance and Data Scarcity in Foundation-Model-Based Histopathology Patch Classification

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Thesis Proposal

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1 Motivation

Pathology foundation models provide transferable representations for histopathology patch classification (Chen et al., 2024; Vorontsov et al., 2024). A classifier can then learn a diagnostic task from these representations without retraining the large model. Public benchmarks compare the models on clinical tasks (Campanella et al., 2025), but offer less guidance on a practical question: **how well do foundation-model-based classifiers cope with class imbalance and scarce training data, and what can be done when they do not?** This matters because expert annotation is expensive (Lu et al., 2021), so labeled data are often scarce, especially for rare cancers.

A classifier is judged here by two properties. *Discrimination* is its ability to assign the correct class. *Calibration* is the reliability of its predicted probabilities: among all patches predicted with 80% confidence, about 80% should be correct (Guo et al., 2017). The two can diverge, since a classifier may pick the right class most of the time and still be systematically overconfident.

Two data limitations are studied. *Class imbalance* means that common diagnoses receive much more training data than rare ones; this can harm both discrimination and calibration (Mosquera et al., 2024). *Data scarcity* means having too little labeled evidence overall, either too few patches or too few independent patients. These are different limits: more patches from one patient may add tissue detail without representing the variation between patients.

The study uses foundation models as backbones and examines class imbalance, data scarcity, and whether the choice of backbone changes their effects. The aim is to explain what annotation effort should target (rare classes, more tissue per patient, or additional patients) and what an algorithm can compensate for instead.

2 Research Questions

Both class imbalance and data scarcity are examined through three questions: how much performance they cost, what causes the loss, and how effectively it can be mitigated. This progression connects the measured effects to explanations and practical remedies (Figure 1).

1. **Performance impact:** How do class imbalance and data scarcity affect discrimination and calibration across tasks and classes, and how do patch count and patient count differ in their effects?
2. **Mechanisms:** What causes these performance losses, and why are some classes and tasks more vulnerable than others?
3. **Mitigation:** How effectively do mitigation methods improve performance under class imbalance and data scarcity, and why do some methods work better than others?

3 Methodology

I design controlled experiments on two patch classification tasks. TCGA-UT contains 30 cancer types from different organs (Komura et al., 2022); BRACS contains seven breast-lesion classes that range from normal tissue to invasive carcinoma (Brancati et al., 2022). Every patch is encoded once by a frozen foundation model, Virchow2 (Zimmermann et al., 2024), and a linear classifier (multinomial logistic regression) is trained on these features. To test whether the backbone matters, the experiments are repeated on the same patients and patches with a second encoder, UNI2-h, the successor of UNI (Chen et al., 2024).

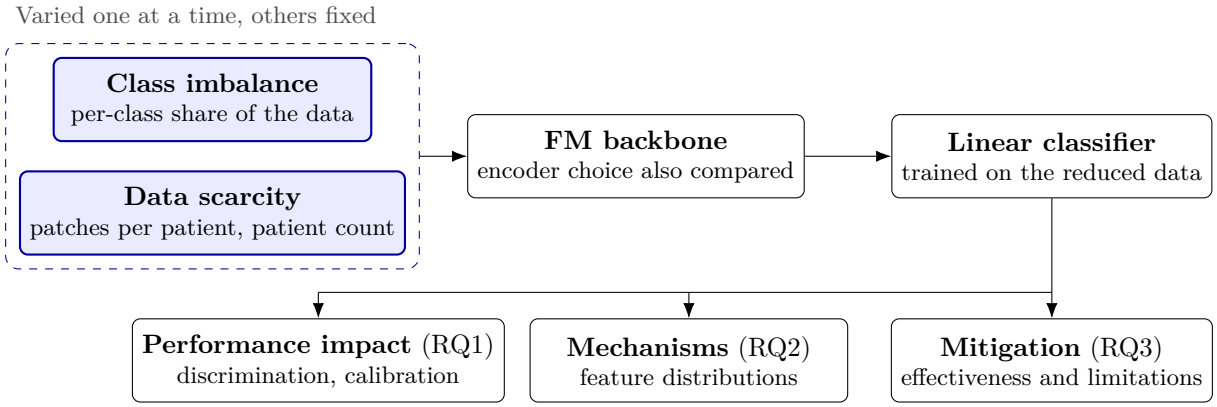


Figure 1: Experimental design. Each data factor (blue) is varied alone while the other stays fixed. The reduced training set is encoded by a frozen foundation model (FM), and the linear classifier trained on it is analyzed for the three research questions.

Patient splits. Each dataset is split by patient, so that all patches of one patient fall into exactly one partition. Within each class, patients are assigned at random to training (about 70%), validation (15%) and testing (15%). Three such splits are drawn with different random seeds. The validation patients serve to select the regularization strength of the classifier and to fit the calibration correction; the test patients are used only for the final evaluation.

For **class imbalance**, the total number of training patches T and the number of patients per class are the same in every setting (20 patients per class on TCGA-UT, 10 on BRACS). What changes is how T is divided among the classes. In the balanced setting every class receives the same number of patches, 32 per patient. In an imbalanced setting frequent classes receive more patches per patient and rare classes fewer (Figure 2). The *imbalance ratio* ρ states how many times more patches the most frequent class receives than the rarest one: $\rho = 1$ is balanced, and $\rho = 100$ is the most extreme ratio tested. Between these two classes, the patch counts decrease geometrically: the class at rank k of K receives a share of T proportional to $\rho^{-(k-1)/(K-1)}$, so each class receives the same fraction fewer patches than the class ranked above it. Each class's patches are divided evenly among its patients. Which class is frequent and which is rare is assigned at random and redrawn ten times per split, so that the result does not depend on one particular choice.

For **data scarcity**, the classes stay balanced and two experiments separate patch count from patient count. In the first, the patients stay the same and each contributes fewer patches. In the second, the total number of patches stays the same but comes from fewer patients, each contributing more patches. Comparing the two shows whether a classifier is limited by the amount of tissue or by the number of independent patients.

Evaluation. Discrimination is measured by *balanced accuracy*: the share of correctly classified test patches is computed per class and then averaged over classes, so that a rare class counts as much as a frequent one; within a class, every patient counts equally regardless of patch count. Calibration is measured by the *expected calibration error* (ECE), the average gap between the confidence of a prediction and its accuracy; lower is better. The effect of an imbalanced or scarce

setting is reported as its difference from the balanced setting on the same patients, in percentage points (pp), with 95% bootstrap intervals over test patients and repetitions.

4 Preliminary Findings

In my initial experiments, I investigated the effects of *class imbalance* on both tasks, the 30 cancer types of TCGA-UT and the seven breast-lesion classes of BRACS. Specifically, the imbalance ratio ρ between the most frequent and the rarest class was varied from 1 to 100 (Figure 2). Each ratio was tested 30 times: three patient splits, each with ten random choices of which class is frequent and which is rare. The numbers below refer to Virchow2 unless stated otherwise. Four findings guide the remaining work:

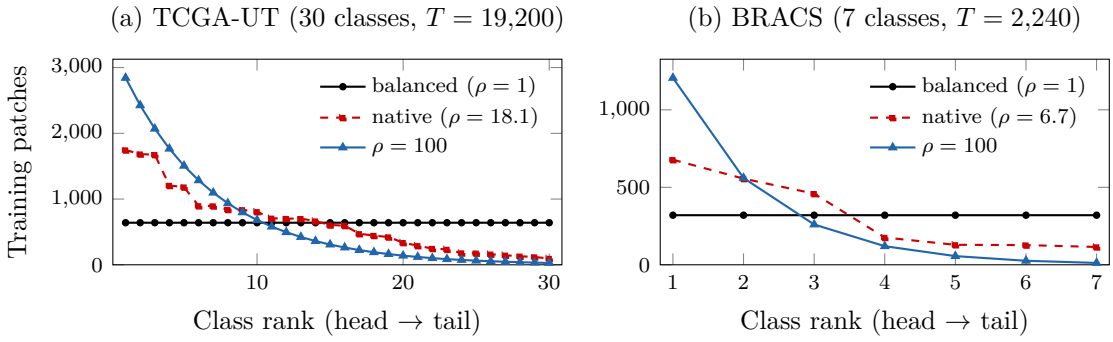


Figure 2: Training class distributions: balanced, native and ratio 100, sorted from most to least frequent. Total patch budget T and patients per class stay fixed; only class shares change.

1. Imbalance costs far more accuracy on BRACS than on TCGA-UT. Figure 3a shows the *damage*: the balanced accuracy lost when the training data change from balanced to imbalanced at ratio ρ. With balanced data, the classifier reached 72.1% balanced accuracy on TCGA-UT and 39.6% on BRACS. TCGA-UT tolerated moderate imbalance: it lost only 0.6 pp at $\rho = 10$ and 2.9 pp at $\rho = 100$. BRACS was more sensitive: it lost 1.2 pp already at $\rho = 2$ and 7.7 pp at $\rho = 100$. The average also hides how hard the rare classes are hit. At $\rho = 100$, the recall of the two rarest BRACS classes fell from 38.7% to 7.3%, so most of their patches were misclassified. A likely reason for the difference between the tasks is how similar their classes are. TCGA-UT separates cancers from different organs, whereas BRACS separates lesions of one organ with similar morphology, which is why BRACS is arguably harder even with balanced data. When two classes look alike, many patches lie close to the boundary between them, and a classifier that leans towards the frequent class then misassigns many patches of the rare one. Finding 2 supports this explanation.

2. Which class is rare matters as much as the imbalance ratio. The same ratio does very different damage depending on which classes end up rare. The losses in finding 1 are averages over random choices of the rare classes. With ρ fixed at 100 on TCGA-UT and only this choice changed, the loss was 1.3 pp in the mildest case and 4.5 pp in the worst (not shown in Figure 3). The worst case was a rare class that is often confused with a frequent one: the classifier then assigned its patches to the frequent class. On BRACS the loss lay between 5.7 pp and 10.0 pp, depending on which class is the rarest. It was highest for invasive carcinoma: this class is recognized best with balanced data, so it has the most to lose. Knowing the ratio is therefore not enough to tell which diagnoses need more labels.

3. A rare class suffers for two reasons, and reweighting addresses only one. A rare class has two problems. It has little *influence*: its few patches carry little weight in training, so the classifier is fitted mainly to the frequent classes. And it shows little *diversity*: at $\rho = 100$ the classifier sees only one or two of its patches per patient. Two additional experiments separated

these problems. In the first, the rare class kept all its patches, but its weight in training was reduced to that of the imbalanced case (low influence only). In the second, it kept only its few patches, but its weight was raised to that of every other class (low diversity only). On BRACS at $\rho = 100$, low influence alone cost 4.5 pp and low diversity alone 3.3 pp; together they explain the 7.7 pp total. On TCGA-UT at $\rho = 100$, low influence alone cost 1.2 pp and low diversity alone 0.5 pp; a further 1.2 pp arose only when both occurred together, which gives the 2.9 pp total. Reweighting gives rare classes more influence but cannot add diversity, so 3.3 pp remain on BRACS. Recovering them needs more patches or patients, or methods that learn a class from few examples.

4. Calibration is easier to repair than discrimination. Imbalance makes the classifier overconfident. When ρ increases from 1 to 100, the ECE rose from 5.5 to 17.8 pp on TCGA-UT and from 12.3 to 35.7 pp on BRACS (solid lines in Figure 3b). *Temperature scaling* corrects this with a single number: all class scores are divided by one constant fitted on the validation patients, which lowers every confidence in the same way (Guo et al., 2017). After this correction the ECE rose only from 1.5 to 2.1 pp on TCGA-UT and from 6.6 to 8.6 pp on BRACS (dashed lines), so 92–96% of the increase caused by imbalance is gone. Dividing all scores by the same constant does not change which class scores highest. The predicted classes, and with them the accuracy losses in Figure 3a, stay exactly the same: a better confidence estimate does not recover a missed diagnosis.

Backbone. Repeating the experiments at $\rho = 10$ and $\rho = 100$ with UNI2-h left these findings intact. This comparison used new random choices of which class is rare, so the Virchow2 losses below differ slightly from those above. On TCGA-UT at $\rho = 100$, UNI2-h lost 2.3 pp (0.9 pp less than Virchow2). On BRACS, UNI2-h lost 7.2 pp and Virchow2 7.9 pp; this difference of 0.7 pp lies within the uncertainty of the estimate, and the recall of the two rarest classes collapsed to about 9% with both. Changing the backbone alone does not solve the problem.

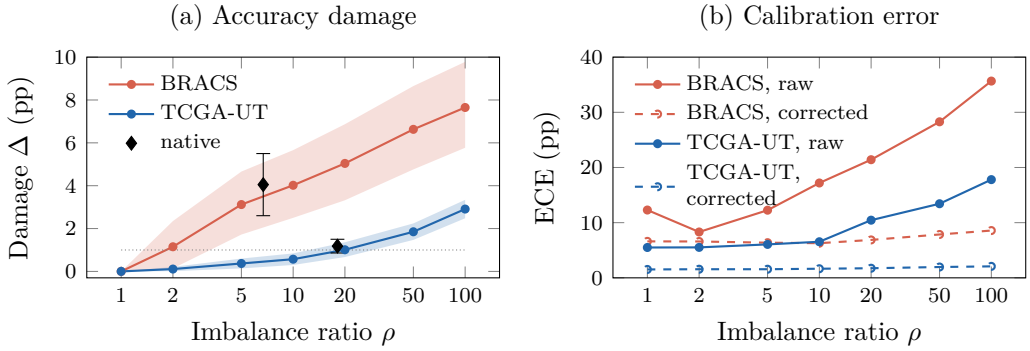


Figure 3: Effect of the imbalance ratio ρ; lower is better in both panels. (a) Balanced accuracy lost relative to balanced training data (percentage points, pp), with 95% intervals; diamonds show the datasets' native class shares and the dotted line marks 1 pp. (b) Expected calibration error (ECE) of the raw predicted probabilities (solid) and after temperature scaling (dashed, "corrected").

5 Next Steps

For my thesis, I plan to extend the imbalance findings to **data scarcity** and then evaluate **mitigation methods for both limitations**. The aim is to establish which methods work well, where they fall short, and why, alongside guidance on where additional labels are most valuable. The remaining work follows two stages:

Data scarcity. In the first stage, I will reduce the number of patches per patient while keeping the patients fixed, and then reduce the number of patients while keeping the total number of patches fixed. Both experiments measure the loss in discrimination and calibration. The analysis then examines whether the variation of tissue within a patient or the diversity between patients limits performance, and from which point additional patches stop helping.

Mitigation. The second stage evaluates methods against class imbalance and data scarcity, including class reweighting, methods that estimate a class from few examples, and temperature scaling. Their effectiveness is compared across tasks and levels of imbalance and scarcity, with an analysis of why some methods succeed while others offer little improvement.

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